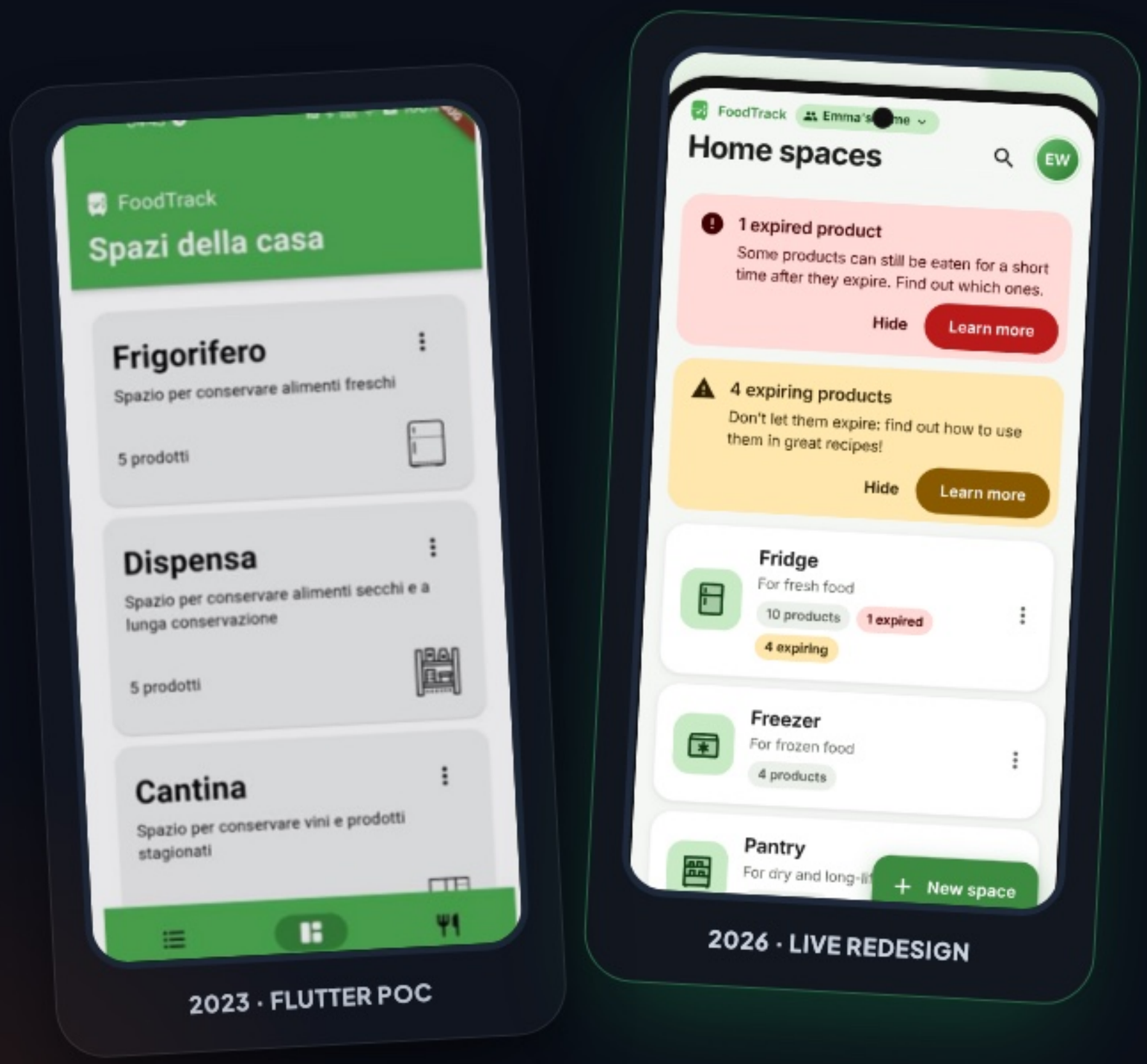
FoodTrack



CASE STUDY · PRODUCT JOURNEY

# From University Exam to Production App

How user testing transformed a simple Flutter prototype with mock data into a full-stack, live application.



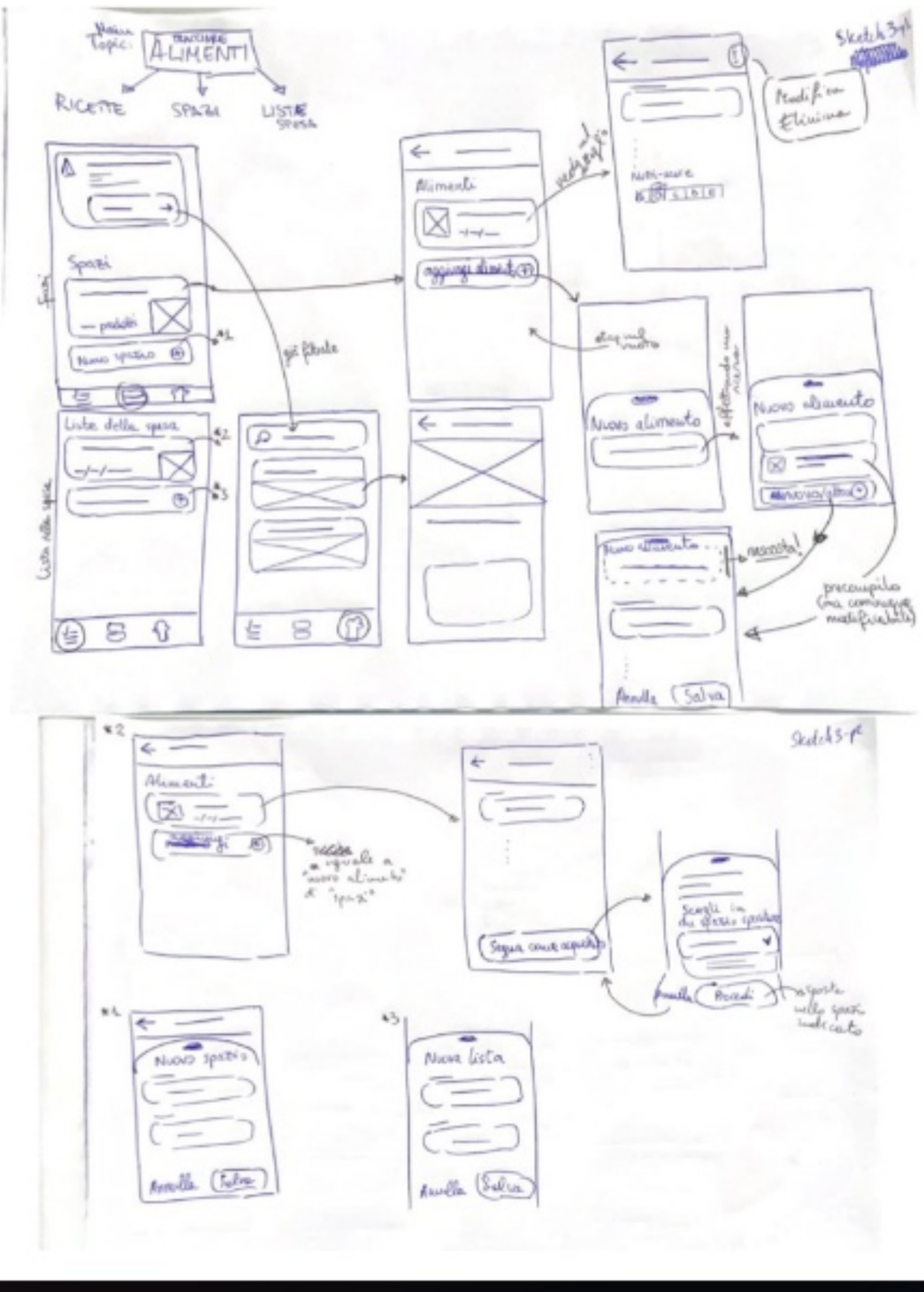

By **Saverio Morelli** · UniUd Project Evolution

Swipe to see the evolution →

PHASE 1 · EARLY IDEATION (2023)

# Paper Sketches & Mental Models

Born in the Mobile Application Design course at Università degli Studi di Udine to solve everyday food waste.




### ✏️ Iterating Mental Models

Initial sketches had a generic "Home" screen that caused confusion. After 3 iterations, we split the app into 3 clear pillars: Spaces, Lists & Recipes.

### 🎯 User Personas

Designed around real needs: busy professionals, university students living alone, and families struggling to track fridge contents.

### 🌱 Nature-Inspired Palette

Mood board #4 was chosen: natural greens and warm accents symbolizing food freshness, balance, and vitality.

PHASE 2 · PROOF OF CONCEPT

# The Academic Flutter Prototype

Building a working mobile PoC for the exam to put our design assumptions to the test.




### 📱 Coded in Flutter

Brought the Figma wireframes to life with a native Flutter UI, responsive bottom navigation, and action sheets.

### ⚠️ Static Mock Data Only

No backend, no database. Everything resided in `mockData.dart` and reset whenever the app closed.

### ✂️ Features Cut Due to Complexity

Multi-user household sync, camera barcode scanning, and recipe ingredients sync had to be shelved for the exam deadline.

PHASE 3 · USABILITY TESTING

# What Real Users Taught Us

Testing the prototype with actual users completely shifted our roadmap and priorities.

### Shared Households, Not Solo Tracking

Pantry management is rarely an individual task. Flatmates and families needed to share the same spaces and shopping lists in real time.

> *"I need to share the shopping list and fridge with my flatmates, or we'll buy duplicates."*

### Frictionless Barcode Scanning

Typing product details manually caused high drop-off. Users needed a quick way to scan items directly with their camera.

> *"Typing expiration dates and product names one by one takes way too much time."*

### Actionable Anti-Waste Recipes

Users didn't just want to see what is expiring; they wanted instant ideas on what to cook with whatever is available right now.

> *"Tell me what dinner I can make with the spinach expiring tomorrow."*

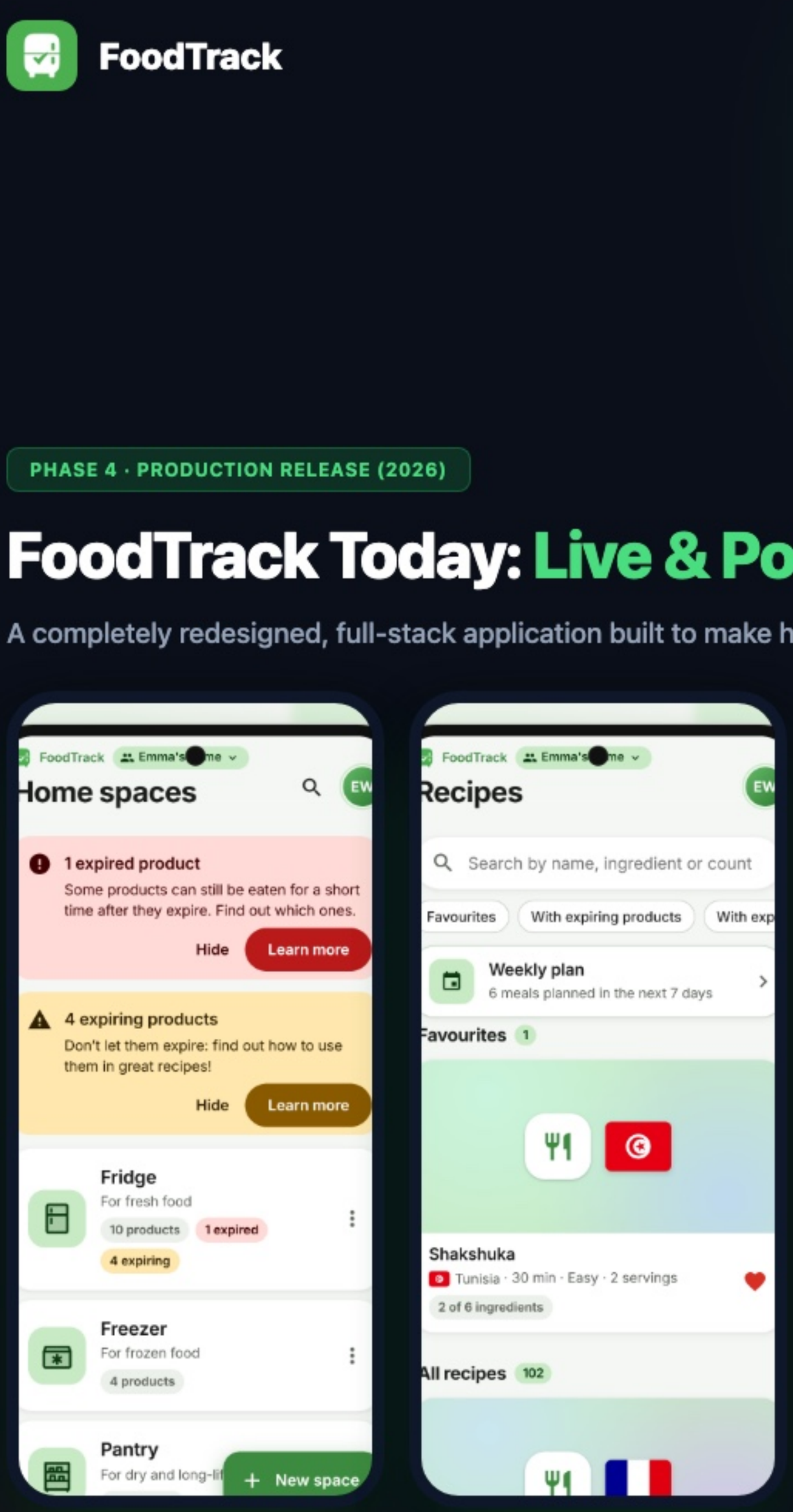
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PHASE 4 · PRODUCTION RELEASE (2026)

# FoodTrack Today: Live & Polished

A completely redesigned, full-stack application built to make home food management effortless.




### ☁️ Real-Time Cloud Sync

Multi-user shared households with instant invite codes and automatic background sync.

### 📦 Integrated Barcode Scanner

Native camera scan connected to open food databases for instant product cataloging.

### 🍲 100+ Global Recipes

Automatically calculates available ingredients: "You have 4 of 6 ingredients ready in your fridge".

### ⚡ Installable PWA & Android App

English & Italian, offline-ready service worker, smart expiry reminders & dark mode.

Explore the live app: **saveriomorelli.com/projects/foodtrack**

**Soon on Google Play** 📲